Model Alpha M14 Failure Analysis Methods Study

HELIX FORGE LAB / p04-full-003708 / RUN R184 / 2026-07-13

ALPHA F1	PRECISION	RECALL	BETA F1
0.873	0.881	0.865	0.846

Decision context

This report records a bounded failure analysis evaluation on HF-Eval-S1/S2 v3.2. The 2,400-record population is synthetic. Alpha M-14 is retained as a research checkpoint, not as a deployment authorization.

Engineering interpretation

Aggregate improvement remains meaningful only with the frozen scoring policy. The review therefore keeps slice-level calibration, dataset lineage, and the comparison with Beta B-07 attached to run R184.

Lineage

Upstream evidence: p04-full-000148 (research/programs/model-alpha/experiments/configs/2026-07-13/model-alpha-m14/seed-stability/tables/model-alpha-m14-seed-stability-s

Slice	Records	Precision	Recall	F1
slice-01	208	0.875	0.873	0.874
slice-02	221	0.892	0.849	0.870
slice-03	198	0.874	0.862	0.868
slice-04	205	0.891	0.875	0.883
slice-05	227	0.873	0.851	0.862